

# Narrative Report in ITEP 413 2



INFORMATION ASSURANCE AND  
SECURITY 2

Prepared by:  
**John Paul R. Ramirez**  
**BSIT-4A**

**Empowering TVET Through  
Cyber Capacity Building**

# Title:

- Oct 9 2025, 2pm, Via Zoom,
- Pasig City, Metro Manila, Philippines.
- Colombo Staff Plan, College, Philippines



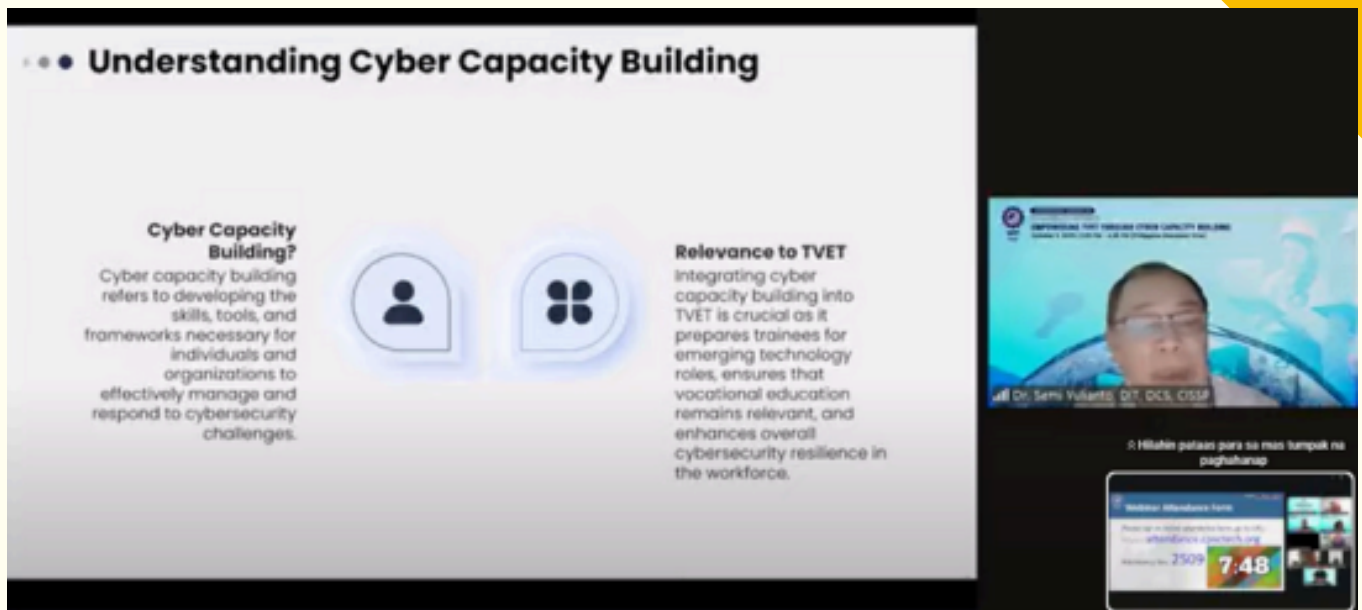
The primary goal is to strengthen Technical and Vocational Education and Training (TVET) institutions and their graduates against emerging cyber threats through the development and implementation of a comprehensive cyber capacity-building program. This initiative seeks to embed core cybersecurity knowledge and best practices into both TVET curricula and institutional operations, ensuring the security of educational infrastructure while equipping graduates with practical skills to support a secure digital workforce. In doing so, it enhances the resilience, relevance, and long-term impact of the TVET sector in the digital era.

## Objectives of the Webinar:

- To provide TVET educators and administrators with practical approaches and resources for effectively incorporating foundational cybersecurity modules and best practices into existing technical and vocational curricula.
- To support TVET leadership and IT personnel in developing and implementing a robust institutional cybersecurity policy framework that safeguards institutional data, student information, and critical network infrastructure.



# Summary of Discussion / Highlights of the Webinar



•The discussion emphasized the pressing need for Technical and Vocational Education and Training (TVET) systems to adopt a comprehensive and proactive approach to cyber capacity building—one that goes beyond basic technology use to focus on institutional resilience and workforce preparedness. Key insights highlighted the importance of embedding essential cybersecurity practices, including cyber hygiene, digital identity protection, and awareness of threats such as phishing and ransomware, directly into TVET curricula to develop “digitally responsible” graduates. Additionally, participants stressed that achieving long-term resilience requires strong leadership to implement effective cybersecurity policy frameworks and ensure the continuous upskilling of educators and IT personnel to safeguard institutional infrastructure and sensitive data in an evolving threat environment.

# Reflections / Personal Insights

My personal reflection on this topic is that genuine digital empowerment in the TVET sector goes far beyond merely providing access to computers and internet connectivity; it requires a fundamental shift in culture that treats security as a core, non-negotiable competency. I came to understand that the challenge is not only the absence of technical resources, but also the lack of a security-aware mindset among both administrators and students. The most striking insight is that by integrating practical, hands-on cybersecurity training directly into vocational programs—whether in network management for IT courses or data protection for logistics and supply chain operations—TVET institutions can make a significant contribution to national cyber resilience while addressing workforce shortages and producing more capable, future-ready graduates.

## Digital Connectivity Data:

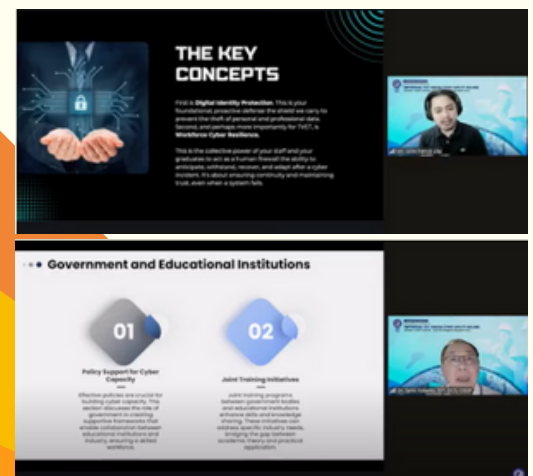
•Digital connectivity continues to expand worldwide, with recent estimates showing that about 74% of the global population—roughly 6 billion people—are Internet users as of 2025, according to the ITU. Despite this progress, substantial inequalities, commonly known as the “digital divide,” remain. These gaps are no longer defined solely by access, but increasingly by differences in connection speed, reliability, and affordability.

## Application / Implication

The cyber capacity building of TVET has profound applications and long-term implications that span institutional policy, curriculum design, and national workforce development. This initiative is the critical link between the Digital Connectivity Data (which reveals both widespread access and significant usage/skill gaps) and the Future of Work (which is increasingly automated, interconnected, and vulnerable to cyber threats).

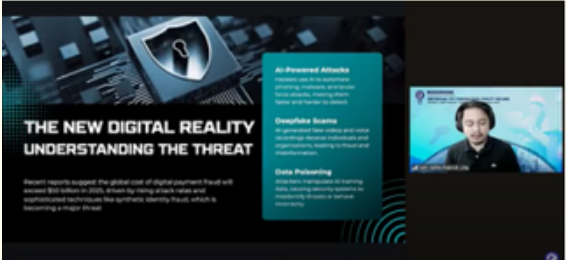
## Discuss here:

- baseline knowledge on data protection, safe online behaviors, and password management.
- TVET institutions themselves are high-value targets for cyber threats due to the sensitive nature of student data and the complexity of their training networks.
- This helps overcome the challenge of rapid technological change and curriculum obsolescence.
- The future success of TVET, and its role in national security, depends on overcoming these challenges through strong public-private partnerships and flexible, adaptive training models.

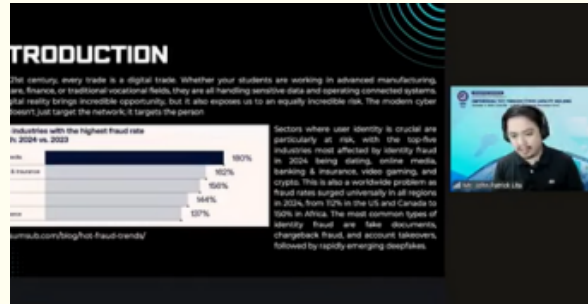


Conclusion:

MSMES Cyber Security



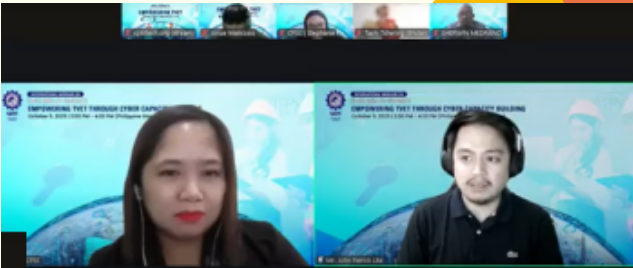
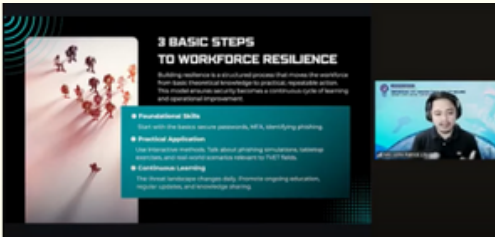
•Micro, Small, and Medium Enterprises (MSMEs) serve as the economic backbone of many countries, but their rapid adoption of digital technologies has often outstripped their cybersecurity readiness, leaving them especially vulnerable to threats such as phishing and ransomware. Unlike large organizations, MSMEs generally lack dedicated cybersecurity budgets, in-house expertise, and well-defined incident response strategies.



•Technical and Vocational Education and Training (TVET) institutions are uniquely positioned to act as an effective platform for building this essential capacity. Their mandate to produce job-ready, skilled workers can be expanded to encompass cyber resilience as a core competency.

•This approach requires strong public-private partnerships in which industry security professionals and MSME associations work closely with TVET institutions to jointly develop curricula and training resources that address immediate, real-world threats. Leveraging virtual training environments and affordable digital tools is crucial to realistically simulate MSME conditions while keeping programs cost-effective. Ultimately, the role of TVET is to foster a security-first mindset within the MSME sector, helping transform vulnerable small enterprises into resilient economic actors capable of confidently operating in the digital economy.

Conclusion:  
Screenshots:



# Certificate:

